

Estimation of Cluster-Specific Causal Effects on Spatially Associated Survival Data Using SoftBART

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The Problem: Treatment Delay in Breast Cancer

Does delaying breast cancer treatment (>90 days) cause worse survival? How does the effect vary across geographic regions?

Breast Cancer Facts

Most diagnosed cancer in US women

2nd leading cause of cancer death

Treatment Delay (TD): diagnosis → first treatment

TD > 90 days ⇒ worse 5-yr survival

Florida Cancer Registry (FCR)

$N = 68,009$ patients, $K = 67$ counties

41.3% right-censored

$Z = 1$ (TD ≤ 90 d) vs. $Z = 0$ (TD > 90 d)

Confounders: Age, HR, Tumor Grade, Biopsy Delay

Key Challenges

Data

➤ **Spatial dependence** — neighboring counties share profiles

➤ **Right-censoring** (41.3%)

➤ **Confounded** treatment (not random)

➤ **Clustered** — patients within counties

Modeling

➤ **Non-linear** treatment–confounder–outcome links

➤ **Unmeasured** county-level confounding

➤ Need **county-** & **patient-specific** effects

➤ No method does spatial + nonparametric + survival

The Gap — No Unified Method Exists

Method	Spatial	Nonparametric	Survival	Causal
IPW / PSW	×	×	×	✓
riAFTBART	×	✓	✓	✓
Spatial Frailty	✓	×	✓	×
PS-LND (Ours)	✓	✓	✓	✓

PS-LND is the first to unify all four in a single Bayesian framework

Causal Estimands

Counterfactual survival: $S_{ij}^{(z)}(t | \mathbf{x}_{ij}) = P(T_{ij}(z) \geq t | \mathbf{x}_{ij}, \text{County} = i)$

Patient-level — CERM (restricted mean gain over $(0, t^*)$):

$$\text{CERM}_{ij}(t^*) = \mathbb{E}[\{T_{ij}(1) \wedge t^*\} - \{T_{ij}(0) \wedge t^*\} | \mathbf{x}_{ij}, i]$$

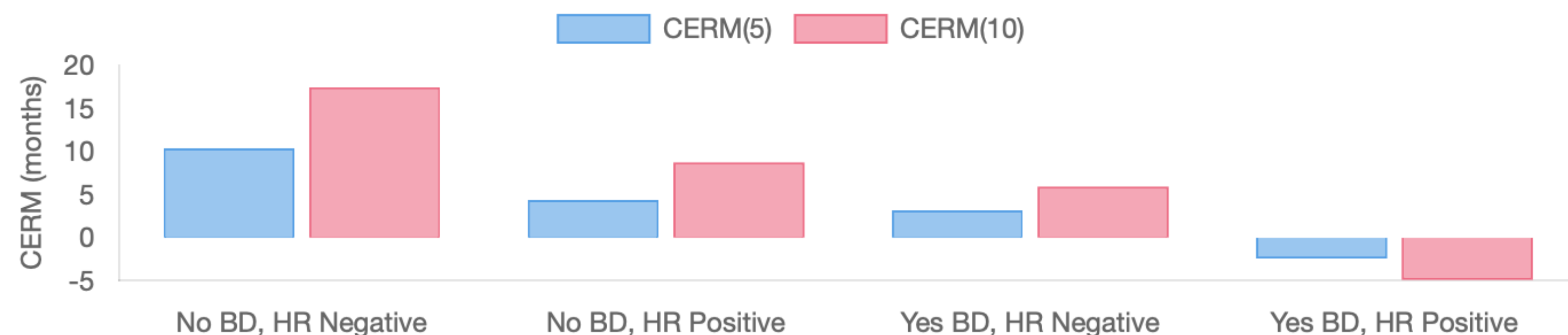
Multi-resolution aggregation:

County: $\text{ACERM}_i(t^*) = \frac{1}{n_i} \sum_j \text{CERM}_{ij}(t^*)$ (also ACERMT, ACERMU)

State: $\text{RATE}(t^*) = \frac{1}{N} \sum_{i,j} \text{CERM}_{ij}(t^*)$ (RATT, RATU)

Survival-prob: $\text{SPTE}_i(t) = S_i^{(1)}(t) - S_i^{(0)}(t)$

Identifiability: SUTVA, Consistency, Unconfoundedness, Positivity, Non-informative censoring



Proposed Method: PS-LND (Soft BART + DAGAR)

Data $(y_{ij}, \delta_{ij}, z_{ij}, \mathbf{x}_{ij}) \Rightarrow$ **Stage 1**: estimate PS $\hat{e}_{ij} \Rightarrow$ **Stage 2**: survival outcome $\Rightarrow$ **causal effects**

Stage 1 — Propensity Score (Spatial Probit-SBART)

$$P(Z_{ij} = 1 | \mathbf{x}_{ij}, V_i) = \Phi(\underbrace{b(\mathbf{x}_{ij})}_{\text{SBART}} + \underbrace{V_i}_{\text{DAGAR}}), \quad \hat{e}_{ij} = E[\Phi\{b(\mathbf{x}_{ij}) + V_i\} | \mathcal{D}_1]$$

$b(\cdot)$: unknown function with Soft BART prior; $\mathbf{V} \sim N(\mathbf{0}, \tau_v^{-1} \mathbf{Q}(\rho_v)^{-1})$ captures **unmeasured spatial confounding**

Stage 2 — Outcome (Log-Normal AFT with BCF structure)

$$\log T_{ij} \sim N\left(\underbrace{\mu(\mathbf{x}_{ij}, \hat{e}_{ij})}_{\text{prognostic SBART}} + \underbrace{\tau(\mathbf{x}_{ij})}_{\text{treatment SBART}} \cdot Z_{ij} + \underbrace{W_i}_{\text{county DAGAR}}, \sigma^2\right)$$

Separate prognostic μ & treatment τ surfaces handle **targeted selection**; $\hat{e}_{ij}$ enters as confounder

$\mathbf{W} \sim N(\mathbf{0}, \tau_w^{-1} \mathbf{Q}(\rho_w)^{-1})$: county random effects; right-censoring via data augmentation

DAGAR spatial prior — interpretable smoothing. For county i with preceding neighbors $\mathcal{N}(i)$:

$$W_i | W_{\mathcal{N}(i)} \sim N\left(\frac{\rho_w}{1+(n_i-1)\rho_w^2} \sum_{j \in \mathcal{N}(i)} W_j, \frac{1+(n_i-1)\rho_w^2}{\tau_w(1-\rho_w^2)}\right)$$

$\rho_w \rightarrow 0$: counties independent; $\rho_w \rightarrow 1$: strong neighbor smoothing

Pairwise correlations stay $\approx \rho_w$ — genuine “borrowing of strength”

Why PS-LND Wins — Key Novelties

- ✓ **Flexible & nonparametric**: Soft BART’s *smooth* sig ✓ **No feedback**: two-stage design isolates PS from mis-specified outcome
- ✓ **Interpretable spatial (DAGAR)**: proper, sparse, positive-definite precision; $\rho_w \in (0, 1)$; scalable
- ✓ **Targeted selection (BCF)**: splits prognostic vs. treatment effect for clean heterogeneity
- ✓ **Doubly robust**: consistent if *either* PS or outcome model is correct
- ✓ **County-specific & multi-level**: full posterior for patient, county & state effects

Simulation: Verifying the Method

$K = 35$ clusters on a 5×7 grid, $N = 1750$, 5 confounders, 150 replications.

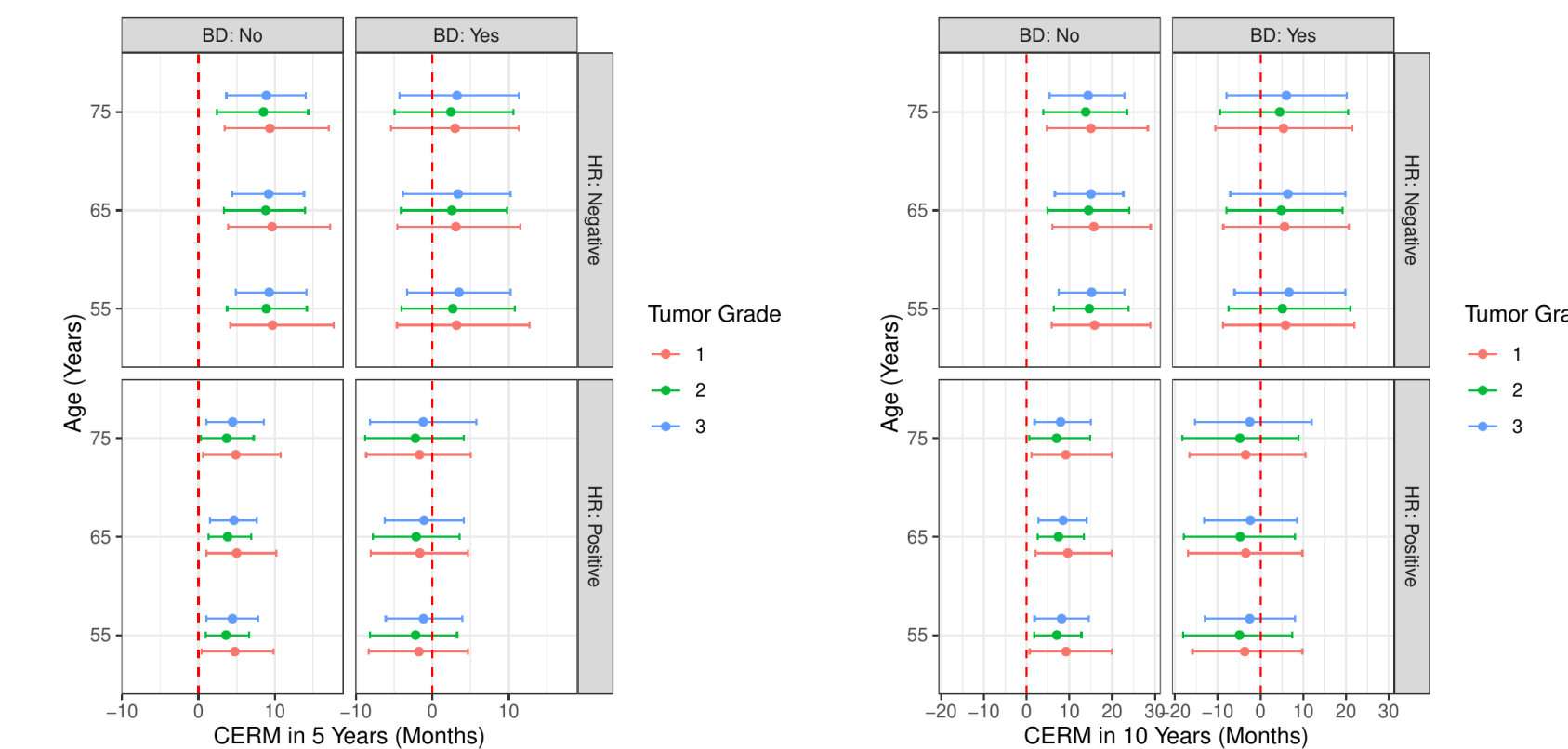
Scenario	What it tests	Data-generating truth
1. Ideal	Correct specification	Log-Normal + DAGAR ($\rho_w = 0.7$); nonlinear main effects + treatment interactions
2. Misspec.	Robustness to model error	Adds covariate $\times$ random-effect interactions
3. Stress	Distribution violation + heavy censoring	Weibull (not log-Normal), 50% censoring, strong confounding

Scenario	Metric	PS-LND	riAFTBART	PSW-Fr.	IPW
1. Ideal	Cov / MAE	0.93–0.97 / 0.01	0.46–0.95 / 0.08–0.58	0.54–0.97 / 0.08–0.50	0.38–0.52 / 0.06
2. Misspec.	Cov / MAE	0.92–0.95 / 0.01	0.77–0.89 / 0.20–0.79	0.64–0.89 / 0.21–0.63	0.42–0.53 / 0.06
3. Stress	Cov / MAE	0.87–0.90 / 0.03	0.55–0.68 / 0.24–0.66	0.76–0.82 / 0.08–0.21	0.42–0.47 / 0.53

PS-LND attains near-nominal coverage & the lowest MAE in every regime — the only method robust to misspecification *and* distributional violation.

Application: Florida Cancer Registry — WA-Distant

Patient-Level Effects (Broward County)



(a) 5-year CERM estimates. (b) 10-year CERM estimates.

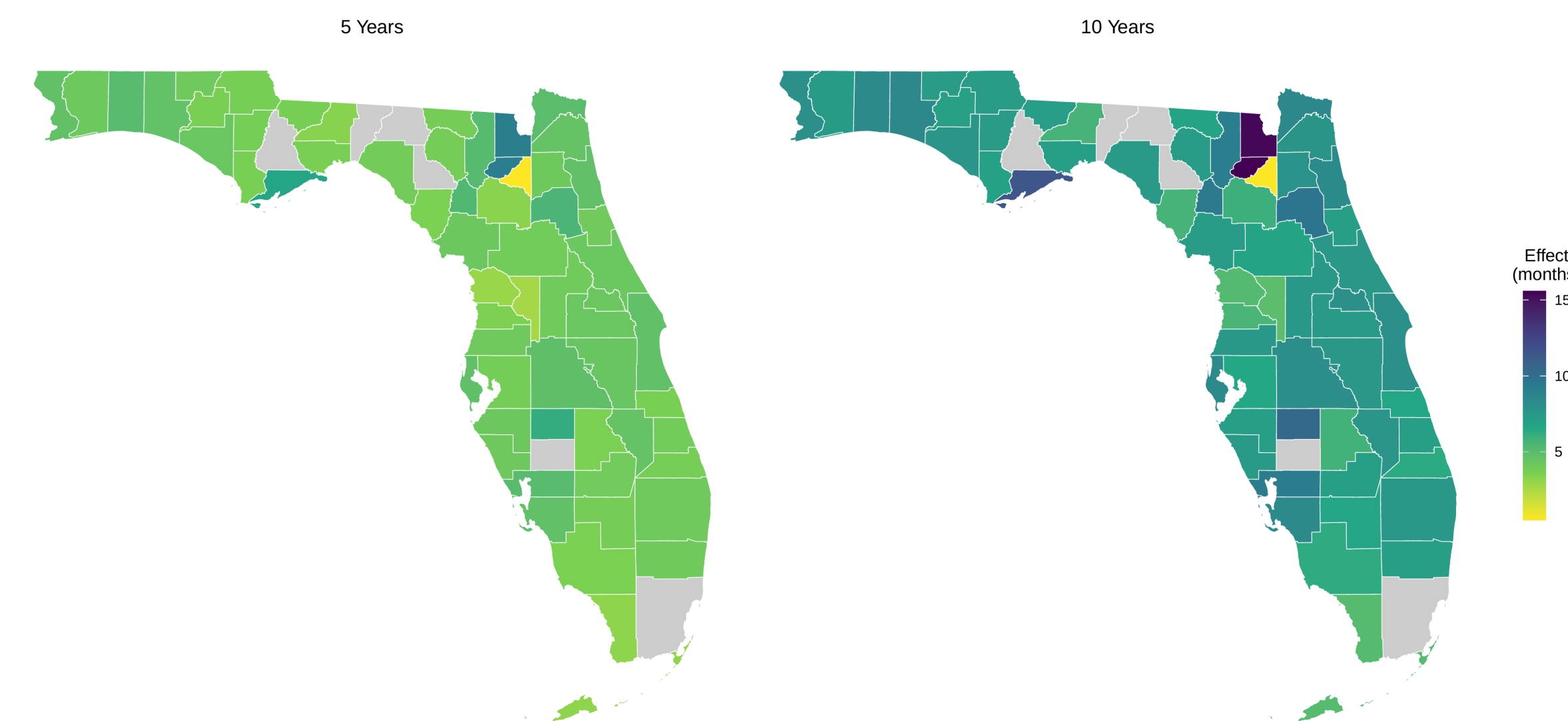
Fig 1: Point and 95% interval estimates of CERM (in months) restricted within intervals of (a) 5 and (b) 10 years for WA patients with Distant-stage cancer in Broward County

HR–, no BD: **13.8–15.9 mo.** gain (10 yr)

HR+, no BD: **6.9–9.6 mo.** gain

With biopsy delay: effect vanishes (CI $\ni 0$)

County-Level Spatial Patterns



Highest: Union (>15 mo.), Baker (>13 mo.)

Near-zero: Bradford (CI $\ni 0$)

State-Wide Treatment Effects

Estimand	N	Est. (mo.)	95% CI
RATE (10 yr)	1994	7.62	(4.42, 10.86)
RATT (10 yr)	687	6.74	(3.09, 10.08)
RATU (10 yr)	1307	8.09	(4.60, 11.42)
RATE (5 yr)	1994	4.43	(2.56, 6.23)

All CIs exclude zero $\Rightarrow$ strong evidence of benefit

RATU > RATT: long-TD patients gain the most

Survival Probability: Peak **11%** survival-probability gain at $t = 2.2$ yr; CI > 0 for all t
Spatial Fit: $\hat{\rho}_w = 0.72$ (strong spatial borrowing); $\hat{\rho}_v = 0.58$ (spatial confounding present)

Policy Takeaway

Short TD $\Rightarrow$ **7.6 mo.** average survival gain statewide

Priority: **HR– patients without biopsy delay**

Target **rural northern counties** for patient-navigation programs

Spatial model reveals **county-level heterogeneity** missed by global estimates